



HK IMO Training 2022-2023 Problem Set 3 Suggested Solutions



Problem 9 (Danube Mathematical Competition 2014 Senior Problem 2). Let k be a positive integer. Consider the set S of all positive integers n satisfying $\lfloor \sqrt{n} \rfloor = k$. Suppose a, b, c, d are elements in S such that $ab = cd$. Prove that $a = c$ or $a = d$.

Solution. Suppose on the contrary that $a \neq c, d$. It is clear that $b \neq c, d$. WLOG assume $a < c \leq d < b$. Let $w = (a, c)$, $a = wx$ and $c = wy$. The equation becomes $xb = yd$. As $(x, y) = 1$, we must have $x \mid d$. Let $d = xz$. This gives $b = yz$.

From $wy = c > a = wx$, we have $y > x$, and hence $y \geq x + 1$. Similarly, from $xz = d > a = wx$, we have $z > w$, and hence $z \geq w + 1$. It follows that

$$b = yz \geq (x + 1)(w + 1) \geq (\sqrt{xw} + 1)^2 = (\sqrt{a} + 1)^2$$

by the Cauchy-Schwarz inequality. This yields $\sqrt{b} \geq \sqrt{a} + 1$, and hence $\lfloor \sqrt{b} \rfloor > \lfloor \sqrt{a} \rfloor$, contradiction. Therefore, we must have $a = c$ or $a = d$. $\square$

Problem 10 (Cyprus TST for JBMO 2021 Test 2 Problem 1). Let x, y, z be positive real numbers satisfying $x^2 + y^2 + z^2 = 3$. Prove that

$$xyz(x + y + z) + 2021 \geq 2024xyz.$$

Solution. By the AM-GM inequality, we have $3 = x^2 + y^2 + z^2 \geq 3\sqrt[3]{x^2y^2z^2}$. This implies $xyz \leq 1$. Also, we have

$$xyz(x + y + z) + 1 = x^2yz + xy^2z + xyz^2 + 1 \geq 4\sqrt[4]{x^4y^4z^4} = 4xyz.$$

Together with $2020 \geq 2020xyz$, we obtain

$$xyz(x + y + z) + 2021 = [xyz(x + y + z) + 1] + 2020 \geq 4xyz + 2020xyz = 2024xyz.$$

Equality holds when $x = y = z = 1$. $\square$

Problem 11 (Czech-Polish-Slovak Match 2015 Problem 2). Let n be a positive integer. Let $\mathcal{F}$ be a family of some subsets of $\{1, 2, \dots, n\}$ such that the following hold: For any $A, B, C \in \mathcal{F}$, one of $(A - B) \cap C$ and $(B - A) \cap C$ is the empty set. Find the largest possible number of elements in $\mathcal{F}$.

Solution. The answer is $n + 1$. For example, when $\mathcal{F}$ contains the empty set and all singleton sets, there are $n + 1$ such sets and it is clear that the condition is satisfied as $(A - B) \cap C$ is always empty.

We prove $|\mathcal{F}| \leq n + 1$ by strong induction on n . When $n = 1$, there are only two subsets of $\{1\}$, so $|\mathcal{F}| \leq 2$ holds. Assume this holds for all smaller cases and consider the case n . If every two sets in $\mathcal{F}$ are disjoint, it is clear that $|\mathcal{F}| \leq n + 1$ since each set except the empty set must contain a distinct element. Now, suppose some pairs of sets in $\mathcal{F}$ are not disjoint. We choose A such that A is a subset of at least two elements in $\mathcal{F}$ and $|A|$ is maximized. Let $A \cup B_1, A \cup B_2, \dots, A \cup B_k$ be all the elements in $\mathcal{F}$ containing A , where each B_i is disjoint with A and every pair of B_i, B_j are disjoint (otherwise $|A|$ is not maximized).

For any other set X in $\mathcal{F}$, there must be $a \in A$ which does not belong to X by construction. Then $(A \cup B_1) - X$ contains a , and hence it is not disjoint with $A \cup B_2$. By the given condition, $X - (A \cup B_1)$ must be disjoint with $A \cup B_2$. In particular, X and B_2 are disjoint (as any common element cannot belong to $A \cup B_1$). Similarly, X is disjoint with every B_i . We remove all elements $A \cup B_i$ from $\mathcal{F}$ except possibly the element A (if one of the B_i 's is the empty set) to form the family $\mathcal{F}'$. Note that all elements in $\mathcal{F}'$ do not contain any elements from $B_1, B_2, \dots, B_k$.

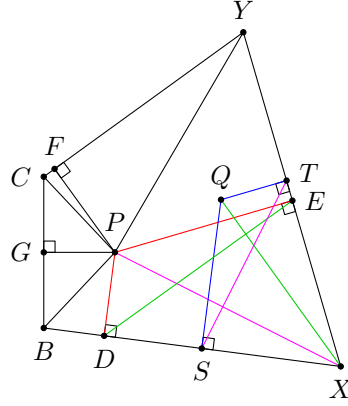
If $A \notin \mathcal{F}$, then we have removed at least k elements (those contained in some B_i). Thus, the elements of $\mathcal{F}'$ are subsets of a finite set with $n - k$ elements. By the inductive hypothesis, $|\mathcal{F}'| \leq n - k + 1$. This yields $|\mathcal{F}| \leq (n - k + 1) + k = n + 1$.

If $A \in \mathcal{F}$, then we have removed at least $k - 1$ elements. Again, $|\mathcal{F}'| \leq n - (k - 1) + 1$, and hence $|\mathcal{F}| \leq (n - k + 2) + (k - 1) = n + 1$.

The proof is complete by induction. $\square$

Problem 12 (Czech-Polish-Slovak Match 2020 Problem 6). Let $\triangle ABC$ be an acute triangle. The tangents at B and C to the circumcircle of $\triangle ABC$ intersect at P . Consider the variable points X and Y lying on the lines AB and AC respectively, such that P lies inside $\triangle AXY$ and $\angle XPY = 2\angle BAC$. Let Z be the reflection of A in the line XY . Prove that the circumcircle of $\triangle XYZ$ passes through a fixed point when X and Y vary.

Solution.



We first prove the following claim.

Claim: Let P be a point in a quadrilateral $BCYX$. Then P has an isogonal conjugate with respect to $BCYX$ if and only if $\angle BPC + \angle XPY = 180^\circ$.

Proof. If P has an isogonal conjugate Q , let D, S be the projections of P, Q onto BX , and let E, T be the projections of P, Q onto XY . Since XP and XQ are isogonals, we have $XQ \perp DE$ and $XP \perp ST$. This yields

$$\angle XST = 90^\circ - \angle DXP = 90^\circ - \angle EXQ = \angle XED.$$

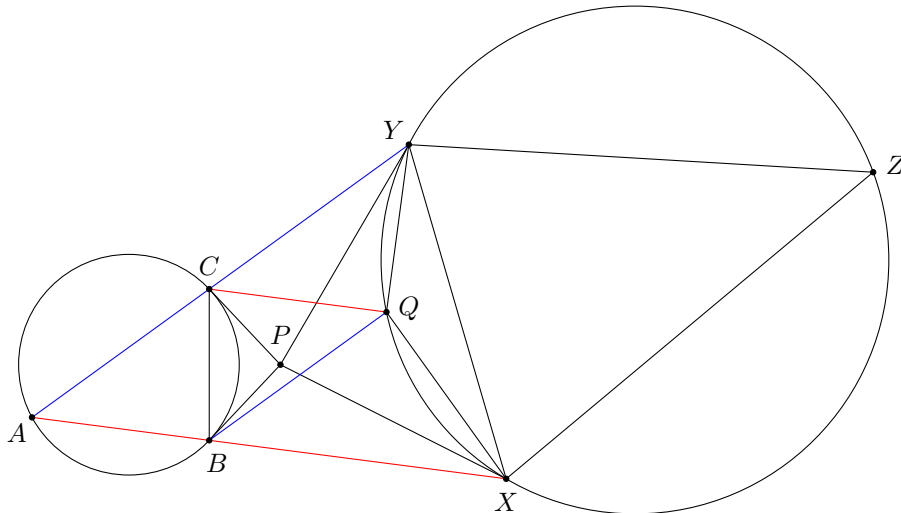
Therefore, D, S, E, T are concyclic. The centre of $(DSET)$ is the intersection of the perpendicular bisectors of DS and ET , which is the midpoint of PQ . Similarly, the projections F, G of P onto CY and BC also lie on the circle whose centre is the midpoint of PQ and passing through D . This shows D, E, F, G are concyclic. This implies

$$\begin{aligned} \angle BPC + \angle XPY &= \angle GPC + \angle GPB + \angle EPX + \angle EPY \\ &= \angle GFC + \angle GDB + \angle EDX + \angle EFY \\ &= 180^\circ - \angle GFE + 180^\circ - \angle GDE \\ &= 180^\circ. \end{aligned}$$

Conversely, if $\angle BPC + \angle XPY = 180^\circ$, we see from above that $\angle GFE + \angle GDE = 180^\circ$. This shows D, E, F, G are concyclic. Let Q be the reflection of P about the centre of $(DEFG)$, and let S, T be the projections of Q onto BX, XY . Then S, T lie on $(DEFG)$. It follows that

$$\angle DXP = 90^\circ - \angle DPX = 90^\circ - \angle DEX = 90^\circ - \angle TSX = 90^\circ - \angle TQX = \angle TXQ.$$

This shows XP and XQ are isogonals. Similarly, Q is the isogonal conjugate of P . □



We return to the original problem. Since

$$\angle BPC + \angle XPY = 180^\circ - \angle PBC - \angle PCB + 2\angle BAC = 180^\circ,$$

P has an isogonal conjugate Q with respect to $CBXY$ by the claim. As $\angle QBX = \angle PBC = \angle BAC$, we have $BQ \parallel AC$. Similarly, $CQ \parallel AB$. This shows $QCAB$ is a parallelogram, and hence Q does not depend on X and Y . Again by the claim, as Q has an isogonal conjugate, we have $\angle BQC + \angle XQY = 180^\circ$. This implies

$$\angle XQY = 180^\circ - \angle BQC = 180^\circ - \angle BAC = 180^\circ - \angle XZY.$$

So Q always lies on (XYZ) , and we are done. □